

Eduardo Hernández Valdez

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EDUCATION

ITESM Av. Eugenio Garza Sada 2501 Sur, Tecnológico, 64700 Monterrey, N.L. Mexico
Engineering in Robotics and Digital Systems; Avg 92/100 - GPA 3.6 August 2022 - June 2026

RESEARCH EXPERIENCE

Joint Robotics Laboratory (JRL) @ CNRS-AIST Tsukuba, Japan
*Research Intern - Humanoid Robotics, targeting **RA-L** and **Humanoids*** Aug 2025 - Feb 2026

- **GPU-accelerated** sensors for robot simulation with **MuJoCo** and **JAX**
- **Tactile-guided reinforcement learning** policy development for **dexterous manipulation**
- **Hardware & software co-design** of tactile matrixes for dexterous humanoid **robot teleoperation**
- Microcontroller interfacing **PCB design** for reading 300+ analog signals and transmitting them over UDP

FLOWMAS: Multimodal Flow Matching for Trajectory Generation Mty, Mexico
Primary Author - CS Dept. @ ITESM - Under revision for System Integration 2026 Sep 2025

- Designed **multimodal CNF model** with **spatial memory** for RGBD-conditioned trajectory synthesis

Self Driving Vehicle @ VantTec Mty, Mexico
Computer Vision, Embedded and Electronics R&D Team Leader Jan 2023 - June 2025

- End-to-end local planner of **autonomous vehicles using generative flow models - ICRA2025 Poster**
- **Embedded** programming and PCB design for **steering wheel and braking pedal control**

Real-time in-situ defect detection for fresh produce using Artificial Intelligence Mty, Mexico
Team Researcher Feb 2024 - June 2024

- Developed CNN architectures to improve accuracy and speed of already existing algorithms using **ONNX**

Unmanned Surface Vehicle @ VantTec Mty, Mexico
Perception R&D Team Leader - Roboat 2024 and 2025 Aug 2022 - February 2025

- On-device object detection inference framework using **TensorRT** with **stereo triangulation** for 3D localization
- Configured **Linux systems, Docker environments, and hardware drivers** for our vehicle's infrastructure

PROJECTS

Embedded EKF for Pose Estimation - Drone interfacing MCU running **estimating filters on sensor readings**
Camera Calibration - CUDA accelerated camera extrinsic and intrinsic **calibration** for lab showoff
Tensorgha - CNN interfacing with Gameboy Advance games. (Self-driving Mario Kart)
ML Teaching - Designed and delivered **machine learning curriculum** for high school students

WORK EXPERIENCE

Computer Vision Intern - Advanced Solutions Lab @ AMI Automation May 2025 - November 2025

- Integration, research and **SIMD** optimization of computer vision algorithms for the steel **industry**

Freelance General Purpose Programming - fiverr.com/eduarch42 Summer 2019, 2020, 2021, 2022

- Developed solutions in **Data Science**, Backend Development, GUI Applications, with an overall 4.9/5 rating

AWARDS & ACHIEVEMENTS

Xignux Challenge 2024 - Top-3 National - Technology **entrepreneurship proposal** with a focus on sustainability
HackMTY 2022 Hackathon - 1st Place - Hardware box designed to store online identity & cryptocurrencies
International Collegiate Programming Contest – Top 7 within ITESM MTY.
International Science Competition @ ITESM – Finalist in Computer Science category.

SKILLS

Programming: *Proficient;* Python, C, Go. *Intermediate;* C++, JS, Matlab, Rust. *Beginner;* CUDA, Verilog

Technologies/Software: Linux, Docker, ROS[1,2], PyTorch, JAX, Tensorflow, TF-Lite, Keras, TensorRT, ONNX, Numpy, MuJoCo, GDB, GCC, Clang, (C)Make, Git,, KiCAD, FreeCAD, OpenCV, Hydra, OmegaConf

Languages: Spanish (Native), English (B2 - PTEG), German (Elementary), Japanese (Learning)